

RESULTS: KL REGULARIZATION AGAINST AN EMA REFERENCE

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1. SETUP

We train Qwen3-4B-Instruct-2507 with TBA on Countdown using asynchronous rollouts ($\Delta = 10$), $\alpha = 0.9$ for the EMA reference, $\beta = 0.005$, $\text{lr} = 10^{-6}$, $K = 16$ samples per problem, sequence length 4096, 720 training steps, evaluation every 20 steps.

Per-token KL.. Two choices for the per-token KL contribution to the advantage:

- *Exact KL*: $\log T_n(y) - \log E_n(y)$, where E_n is the EMA reference policy.
- *Approx KL*: $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$, the first-order surrogate derived in `main.tex` (eq. 2.10). This avoids a forward pass through E_n at training time.

Centering. For each rollout i , the per-token KL is summed over masked tokens to give a per-rollout KL term $K^{(i)} = \sum_t \text{kl_per_token}(y_t^{(i)})$. The advantage is then adjusted by

$$(1) \quad \text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta \cdot (K^{(i)} - \bar{K}),$$

where $\bar{K}$ is a centering term. Two choices:

- *K-centered*: $\bar{K} = \frac{1}{K} \sum_{j=1}^K K^{(j)}$, the within-group mean of $\{K^{(j)}\}_{j=1}^K$ over the K rollouts sharing a problem.
- *No centering*: $\bar{K} \equiv 0$, so $\text{advantage}^{(i)} \leftarrow \text{advantage}^{(i)} - \beta K^{(i)}$ directly.

2. 2 × 2 MATRIX: EMA REFERENCE

Crossing the two per-token KL choices with the two centering choices yields four cells:

Cell	kl_per_token(y)	Centering term $\bar{K}$
Exact, K-centered	$\log T_n(y) - \log E_n(y)$	$\frac{1}{K} \sum_{j=1}^K K^{(j)}$
Exact, no centering	$\log T_n(y) - \log E_n(y)$	0
Approx, K-centered	$\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$	$\frac{1}{K} \sum_{j=1}^K K^{(j)}$
Approx, no centering	$\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$	0

(In the K-centered rows, $K^{(j)}$ is built from the same row’s kl_per_token.)

Cell	Peak eval (step)	Final eval
Exact, K-centered	0.789 (240)	0.656
Exact, no centering	0.827 (620)	0.821
Approx, K-centered	0.794 (380)	0.548
Approx, no centering	0.831 (520)	0.819

The headline observations:

- (1) Within either KL choice, removing the centering term strictly helps.
- (2) The approx-KL row tracks the exact-KL row closely: peaks differ by 0.005 and finals by 0.002 in the no-centering pair. This is the empirical justification for substituting the first-order surrogate (no extra forward) for the exact EMA KL.
- (3) The K-centered cells exhibit the same kind of late-run instability under both per-token KL choices, suggesting the failure mode is centering-related rather than KL-formula-related.

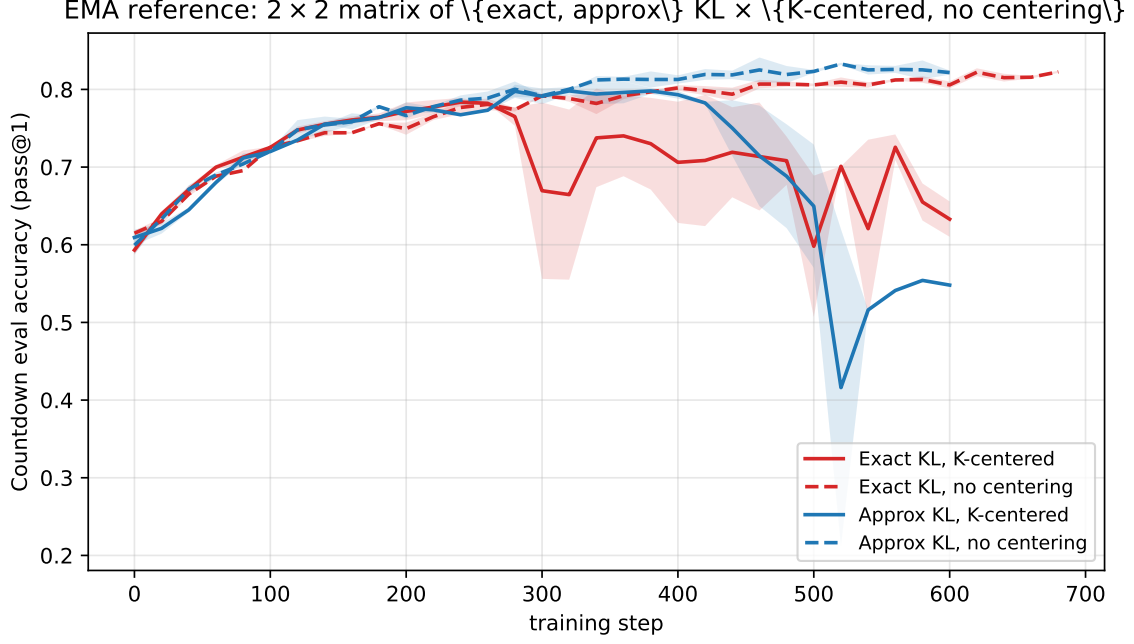


FIGURE 1. Eval accuracy on Countdown for the four cells of the EMA-reference matrix. Red: exact KL. Blue: approx KL. Solid: K-centered. Dashed: no centering. The two no-centering cells (dashed) plateau above 0.80 and peak above 0.82; the two K-centered cells (solid) peak earlier and degrade after step ≈ 300 .

3. RESET REFERENCE (BASELINE)

Before the EMA work, the same form of objective was tested with a periodic-reset reference policy R (hard copy from the training policy every 50 rollout steps). The advantage update (1) is unchanged, but per-token KL is computed against R instead of E_n , and the source for $\log T(\cdot)$ becomes configurable. Let T_n denote the live training-policy forward at the current step and $T_{n-\Delta}$ the inference snapshot used during rollout generation. The per-sample per-token KL is

$$(2) \quad \log T_{\text{src}_1}(y) - \log R(y),$$

and the K-group mean term substitutes a (potentially different) source T_{src_2} :

$$(3) \quad \bar{K} = \frac{1}{K} \sum_{i=1}^K \sum_t \left(\log T_{\text{src}_2}(y_t^{(i)}) - \log R(y_t^{(i)}) \right),$$

where each $\text{src}_j \in \{T_n, T_{n-\Delta}\}$. We test three combinations, each with importance sampling on or off, all with K-centering:

Variant (short)	Per-sample $\log T_{\text{src}_1}$	K-group mean $\log T_{\text{src}_2}$
T/I	$\log T_n$ (live train forward)	$\log T_{n-\Delta}$ (inference snapshot)
T/T	$\log T_n$	$\log T_n$
I/I	$\log T_{n-\Delta}$	$\log T_{n-\Delta}$

The first slot is the per-sample $\log T$; the second is the K-mean $\log T$. Historically these mapped to “klMeanInference”, “klMeanTrain”, and “klAllInf” respectively.

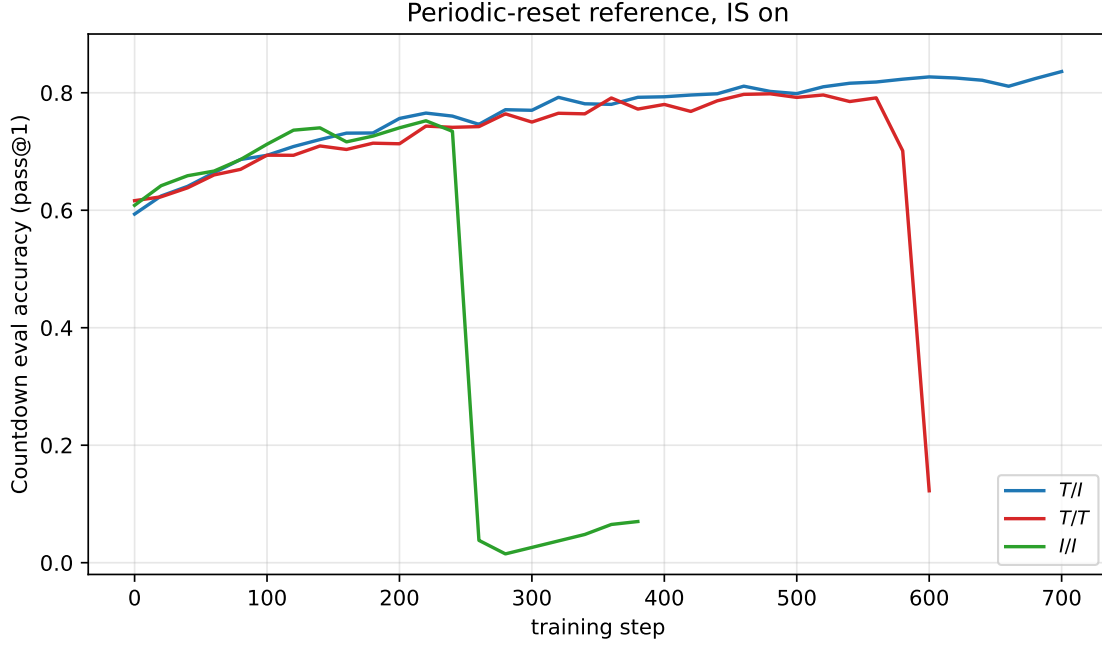


FIGURE 2. Eval accuracy on Countdown with periodic-reset reference and importance sampling on. T/T (red) collapses late in training; I/I (green) is unstable; only T/I (blue) trains stably to high accuracy.

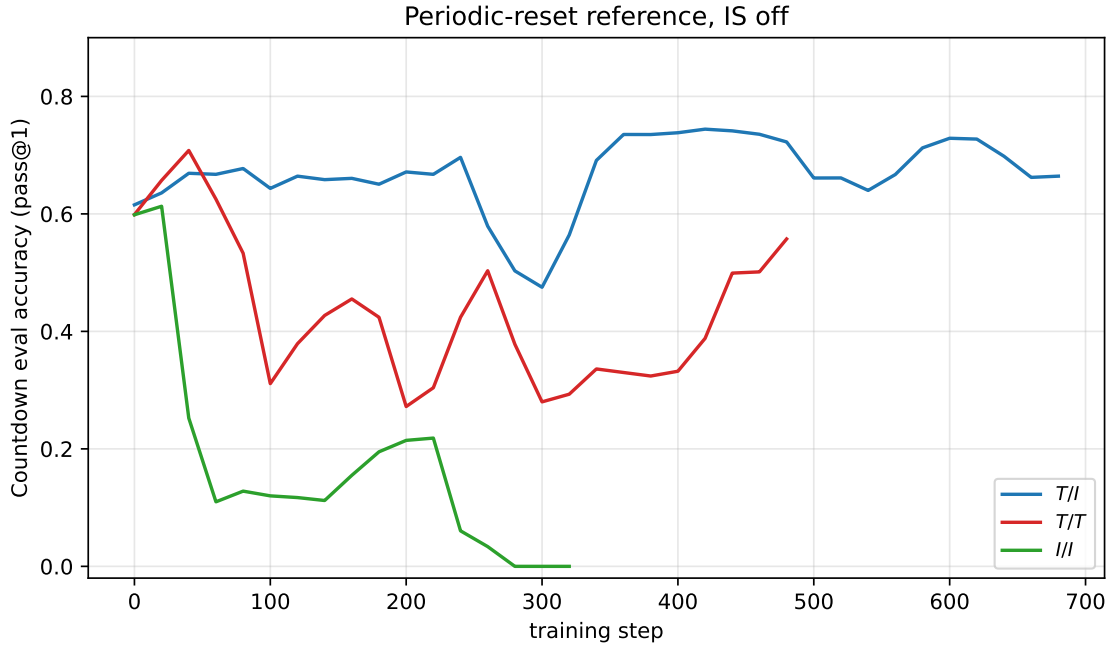


FIGURE 3. Eval accuracy on Countdown with periodic-reset reference and importance sampling off. The same ordering holds — T/I best, T/T collapses, I/I gets stuck near zero — and all three are weaker than their IS-on counterparts.

Run	Peak eval (step)	Final eval
T/I	0.836 (700)	0.836
T/T	0.798 (480)	0.122
I/I	0.752 (220)	0.070
T/I , no IS	0.744 (680)	0.664
T/T , no IS	0.708 (40)	0.557
I/I , no IS	0.613 (20)	0.000

The T/T collapse under reset reference mirrors the K-centered instability in the EMA matrix: in both cases, the per-sample and K-mean KL terms are computed from the *same* drifting source T_n , so the centering subtraction tracks the drift rather than canceling it.

4. APPROXIMATION ACCURACY

We track the per-token error of the first-order surrogate against the exact EMA KL inside `tba_loss`: **MAE** (mean absolute error over masked tokens) and **bias** (signed mean). Figure 4 compares the two no-centering cells (where the loss path differs but both have the same telemetry).

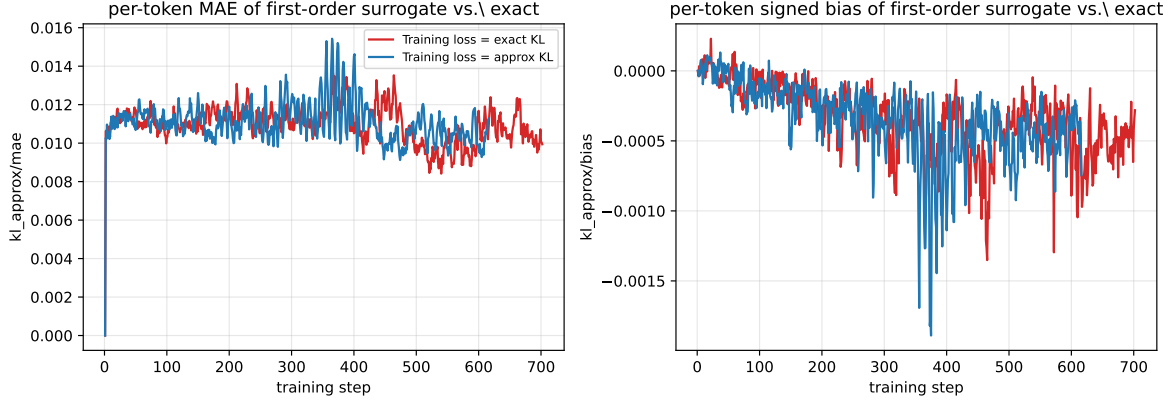


FIGURE 4. Per-token MAE (left) and signed bias (right) of the first-order surrogate $\frac{\alpha}{\Delta(1-\alpha)} (\log T_n(y) - \log T_{n-\Delta}(y))$ against the exact $\log T_n(y) - \log E_n(y)$, averaged over masked tokens within each rollout. Both runs are no-centering EMA-reference cells; they differ only in which formula enters the loss (red: exact KL; blue: approx KL).

MAE stays in the low 10^{-2} band throughout training, and the signed bias is at most a few parts in 10^3 . The approximation is empirically faithful, regardless of whether it is used in the loss.

Across the α -sweep. The same metric is logged for every α run. The intrinsic MAE scales with c_α , since the exact KL grows like $c_\alpha \cdot (\log T_n - \log T_{n-\Delta})$. We therefore also track a rescaled quantity, $(c_{0.9}/c_\alpha) \text{KL}_{\text{exact}} - c_{0.9} \text{diff}$, whose magnitude should be roughly α -invariant if the first-order surrogate has the same fractional quality across α .

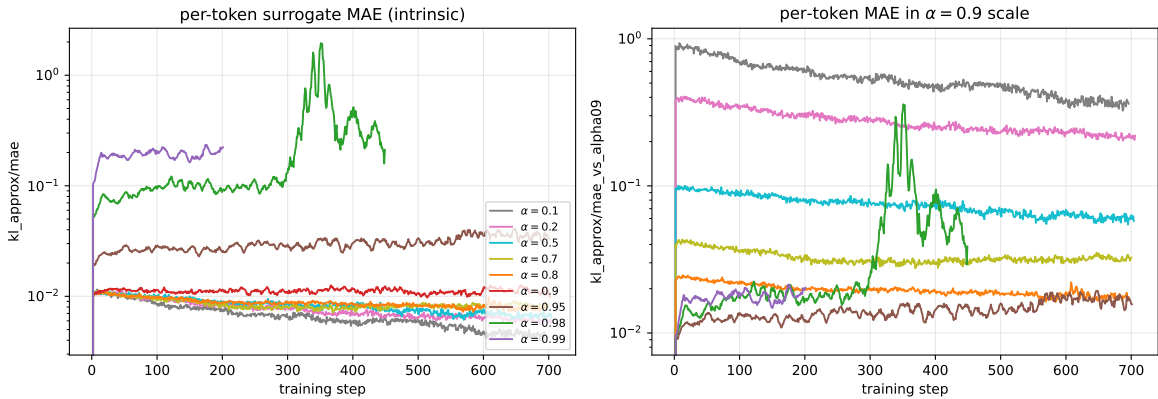


FIGURE 5. Per-token surrogate MAE across the α -sweep. Left: intrinsic MAE (scales with c_α). Right: MAE after rescaling to the $\alpha = 0.9$ surrogate; comparable across α values.

5. ABLATIONS

We probe three knobs in the no-centering / K-centered EMA cells. Some of the runs cited here are still in progress, so trend lines for the newest variants may be short.

5.1. β ablation on the K-centered cell (A). Cell A (EMA exact + klMeanTrain centering) showed a late-run collapse around step ~ 300 at the baseline $\beta = 0.005$. Does scaling the KL penalty up rescue it?

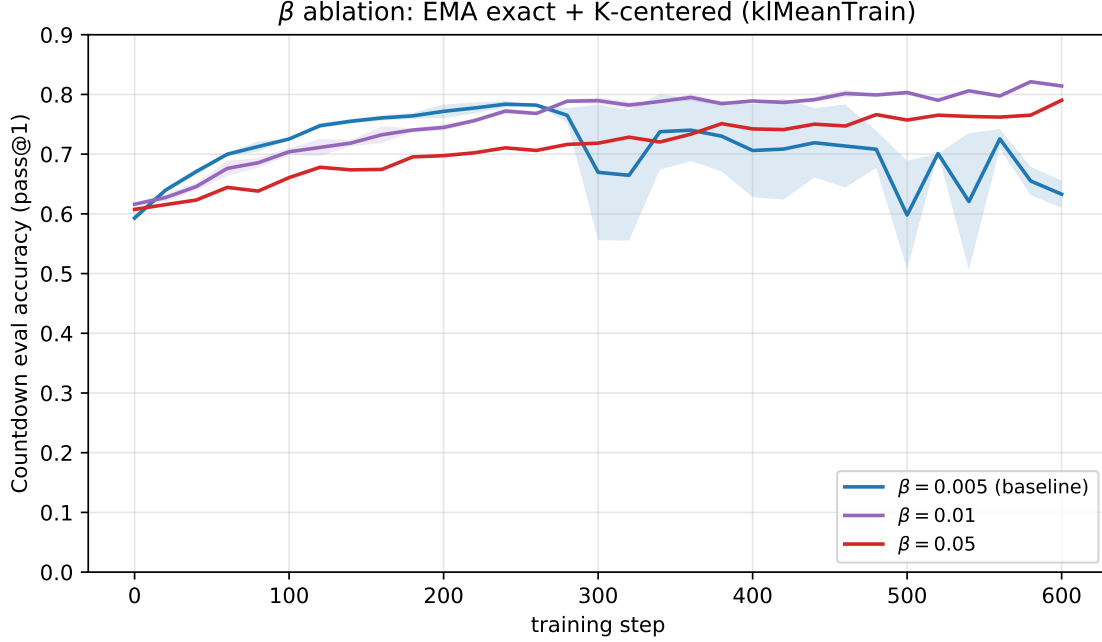


FIGURE 6. β ablation on K-centered exact KL (cell A). Larger β trades off learning speed against the K-centered instability.

5.2. Importance sampling ablation on no-centering. The no-centering cells B and D (exact and approx) were the top-performing matrix entries at $\beta = 0.005$ with IS on. Turning IS off keeps the same conclusion that no centering is best, but at lower absolute accuracy and with late-run instability.

5.3. α ablation on no-centering exact. Sweep the EMA decay $\alpha \in \{0.1, 0.2, 0.5, 0.7, 0.8, 0.9, 0.95, 0.98, 0.99\}$ while holding the effective regularization strength constant ($\beta \cdot c_\alpha = 0.0045$, where $c_\alpha = \alpha/(\Delta(1 - \alpha))$).

5.4. β ablation on no-centering, IS off (B-noIS). The no-centering exact cell with IS disabled also shows late-run instability at the baseline $\beta = 0.005$. Does scaling β up help here too?

5.5. β ablation on reset reference, T/T . The reset-reference T/T variant collapsed late at the baseline $\beta = 0.005$. Does scaling β up rescue the collapse, as it might for the analogous EMA-reference K-centered cell?

5.6. β ablation on reset reference, I/I . The I/I variant uses the inference snapshot for both per-sample and K-mean KL. Since per-sample KL uses inference logprobs, we modified the reference reset to copy from the inference snapshot rather than the live train model (so R consistently lags T_{inf}). We then probe whether raising β helps the I/I cell.

5.7. β ablation on no-centering, approx KL (cell D). We also sweep β for the approx-KL no-centering cell to test the regularization sensitivity of the approx-KL path (cell D, $\alpha = 0.9$).

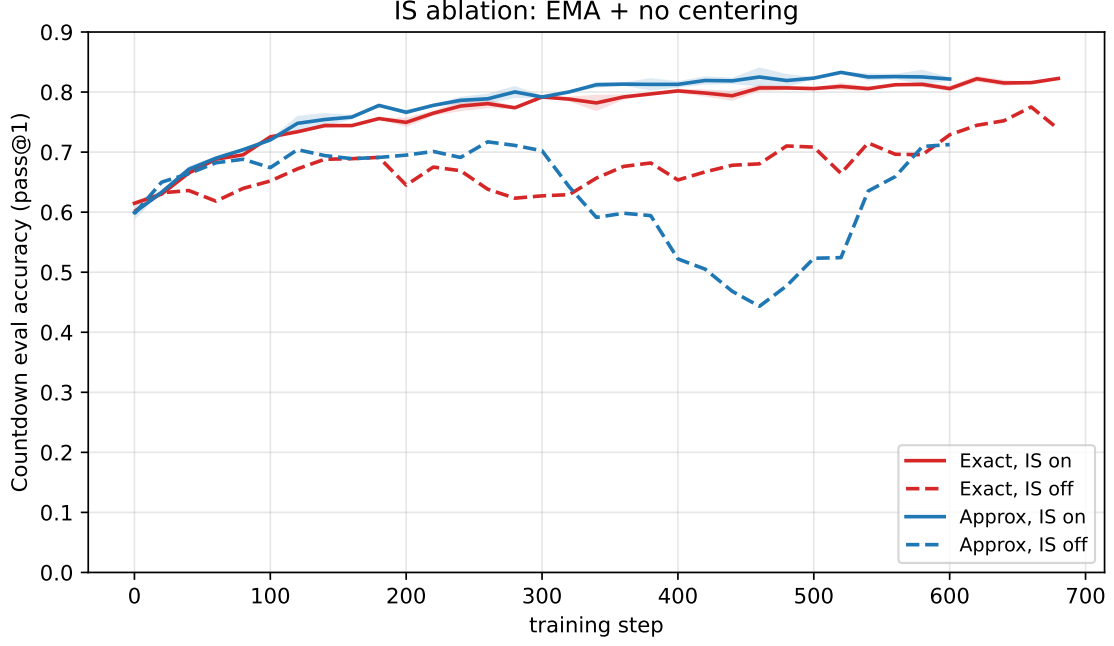


FIGURE 7. IS ablation on no-centering cells. Solid: IS on. Dashed: IS off. Red: exact KL (cell B). Blue: approx KL (cell D).

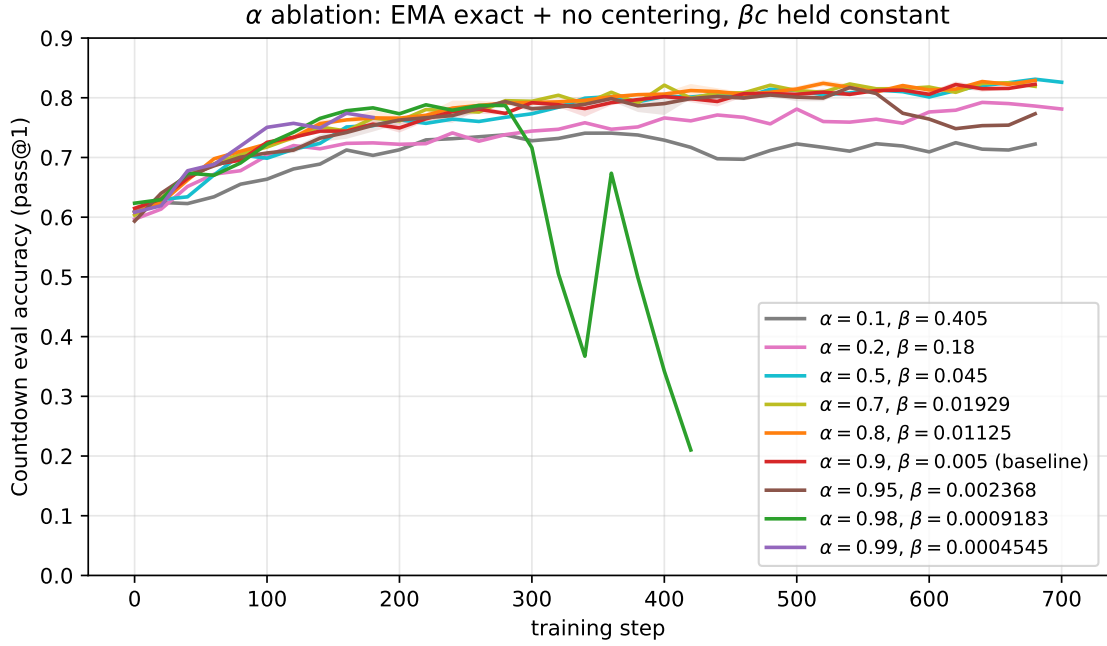
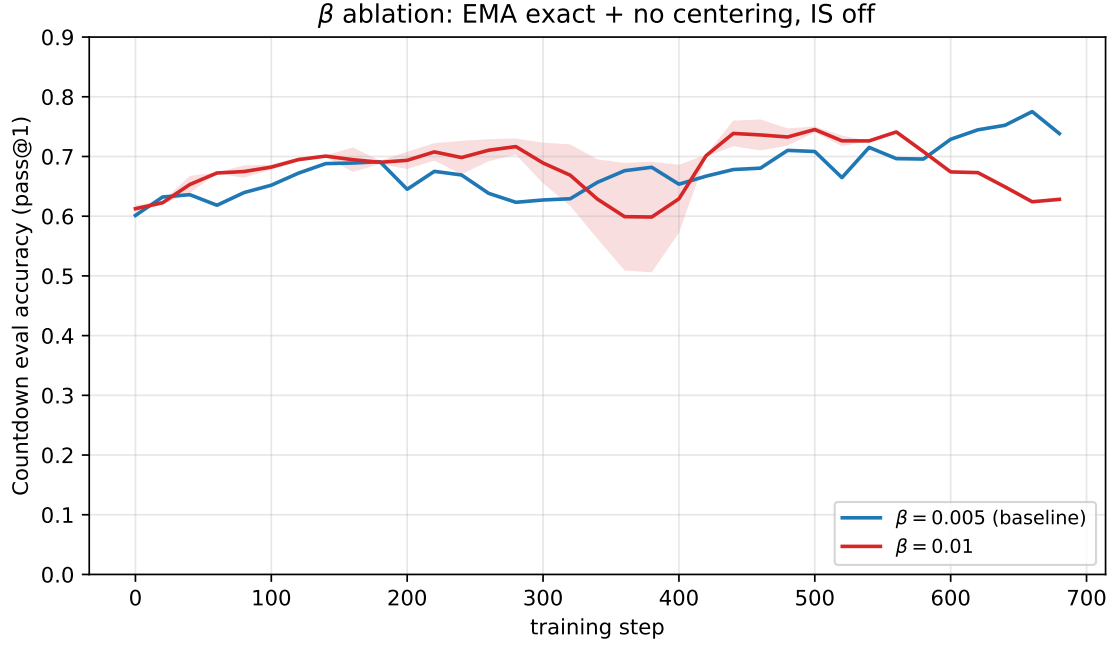


FIGURE 8. α ablation on no-centering exact KL with β scaled to keep $\beta \cdot c_\alpha$ constant. Higher α (slower EMA) becomes unstable: $\alpha = 0.99$ collapses around step ~ 200 , $\alpha = 0.98$ partially collapses around step $\sim 300-400$; $\alpha \in \{0.8, 0.95\}$ match or beat the $\alpha = 0.9$ baseline.

6. CONCLUSION

Removing the K-group centering is the dominant lever: both EMA exact and EMA approx benefit. With centering off, the cheap first-order surrogate matches the exact EMA KL within

FIGURE 9. β ablation on no-centering exact KL with IS off (B-noIS family).FIGURE 10. β ablation on the reset-reference T/T cell (klMeanTrain). Larger β prevents the late-run instability seen at $\beta = 0.005$ (no collapse observed through step ~ 580 at $\beta = 0.01$ or $\beta = 0.05$).

0.005 on peak eval and 0.002 on final eval, so it is a viable drop-in replacement that saves a forward pass through the reference model.

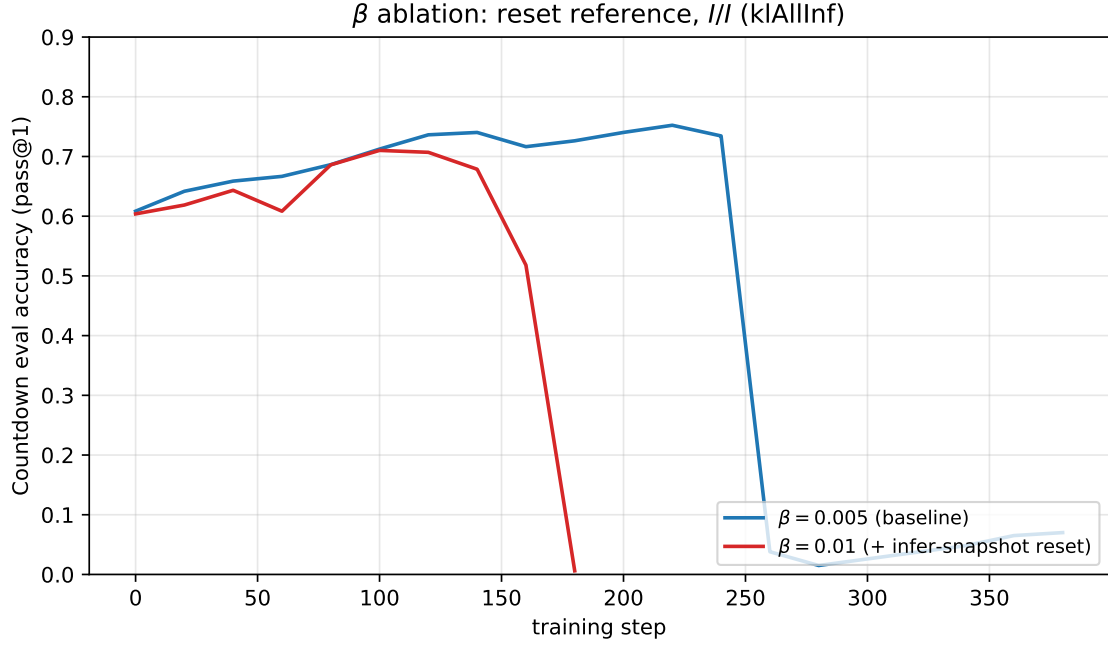


FIGURE 11. β ablation on the reset-reference I/I cell (klAllInf). The $\beta = 0.01$ variant also uses the inference-snapshot reset fix.

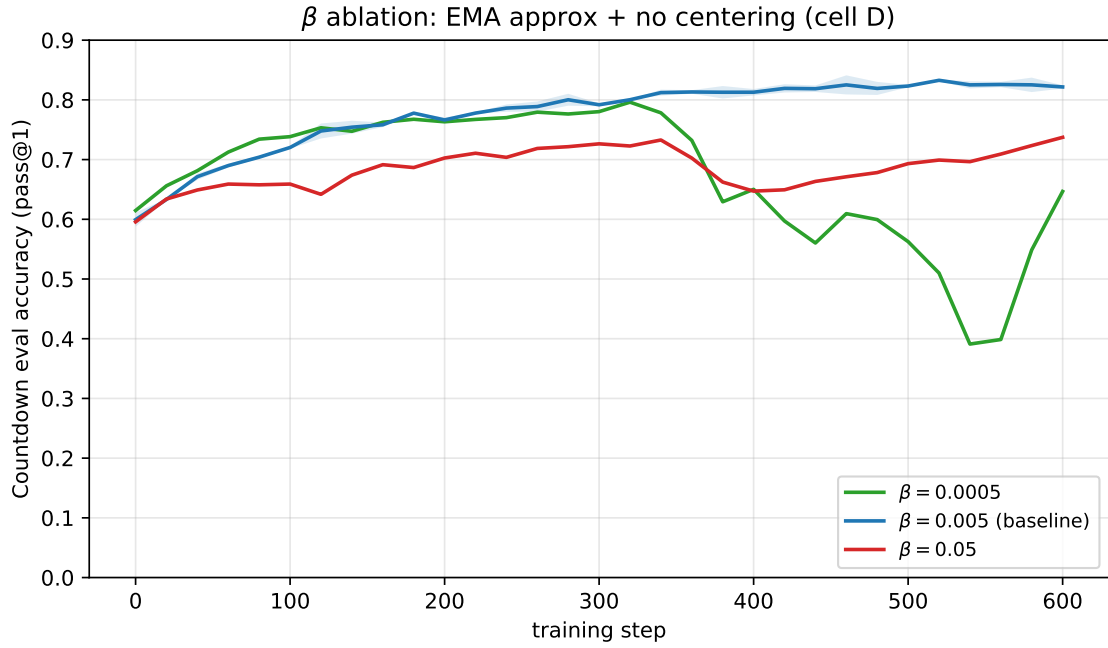


FIGURE 12. β ablation on no-centering approx KL (cell D). Baseline $\beta = 0.005$ vs. $\beta = 0.05$ ($10\times$) and $\beta = 0.0005$ ($0.1\times$). The cell is robust to weakening β (no early-run impact at $\beta = 0.0005$) but slower learning at $\beta = 0.05$.